Stat 380 Final

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Predicting Drowsy Driving: Can we Machine Learning Model to be able to Predict Drowsiness in Drivers?

Introduction

Intuition Behind the Choice of Data Set

- The data set must be large to be able to get a good testing pool.
- The data set has meaningful predictors are practically relevant to our test case.
- The data set contains a drowsy/awake variable
- The data set follows **F.A.I.R.** and **C.A.R.E.** principles.

Ethical Considerations in the Choice of our Data Set

In order to ensure the ethical feasibility of the data set, we also scrutinized our selection to follow the **F.A.I.R.** and **C.A.R.E.** principles as discussed in class.

Since we had obtained our data from the open-source data hub Kaggle, we can ensure that anyone can **find** and **access** this data set through Kaggle's use guidelines, stating for a free distribution of data sets that are published for public use to practice ML or data analysis on. The data is also highly versatile, taking a csv format. This is one of the most **interoperable** formats there are for data sets, especially for our use-case in R, where reading in a csv is already built into the base. Lastly, the data set can be **reused** for multiple types of data analysis, and will always remain relevant for its time period.

In terms of **collective benefit**, the data set contributes to an overall understanding of driving without harming any individuals , due to the fact that it is simply raw data. Secondly, as I had stated the author's name, there is clearly ownership over the data set coming from kaggle as mentioned, showing proper **Authority to Control**. Since proper attributions are maintained, it also falls under the **Responsibility and Ethics** guidelines of the C.A.R.E. principles.

The dataset can be found here.

Cleaning

Aiden, Jordan, Here are some notes for yall. We gotta make sure we handling pulling PNGs and JPEG (not jpg).

Note that the dataset name is Drowsy_datset

Table 1: First 5 NATURAL Train Files:

File
Drowsy_datset/train/NATURAL/1000.png
Drowsy_datset/train/NATURAL/1001.png
Drowsy_datset/train/NATURAL/1002.png
Drowsy_datset/train/NATURAL/1003.jpg_0.png
Drowsy_datset/train/NATURAL/1004.png

Table 2: First 5 DROWSY Train Files:

File
Drowsy_datset/train/DROWSY/1.jpg_0.png
Drowsy_datset/train/DROWSY/1.png
Drowsy_datset/train/DROWSY/10.jpg_0.png
Drowsy_datset/train/DROWSY/10.png
Drowsy_datset/train/DROWSY/100.png

Table 3: First 5 NATURAL Test Files:

File
Drowsy_datset/test/NATURAL/v1013.png
Drowsy_datset/test/NATURAL/w0002.png
Drowsy_datset/test/NATURAL/w0003.png
Drowsy_datset/test/NATURAL/w0004.png
Drowsy_datset/test/NATURAL/w0005.png

Table 4: First 5 DROWSY Test Files:

File
Drowsy_datset/test/NATURAL/v1013.png
Drowsy_datset/test/NATURAL/w0002.png

File
Drowsy_datset/test/NATURAL/w0003.png
Drowsy_datset/test/NATURAL/w0004.png
Drowsy_datset/test/NATURAL/w0005.png

This is showing that some files are PNG and some are JPEG.

```
Drowsy_datset/train/DROWSY/1.jpg_0.png  => PNG check:  TRUE
Drowsy_datset/train/DROWSY/1.png  => PNG check:  FALSE
Drowsy_datset/train/DROWSY/10.jpg_0.png  => PNG check:  TRUE
Drowsy_datset/train/DROWSY/10.png  => PNG check:  FALSE
Drowsy_datset/train/DROWSY/100.png  => PNG check:  FALSE

Drowsy_datset/train/DROWSY/1.jpg_0.png  => JPEG check:  FALSE
Drowsy_datset/train/DROWSY/1.png  => JPEG check:  TRUE
Drowsy_datset/train/DROWSY/10.jpg_0.png  => JPEG check:  FALSE
Drowsy_datset/train/DROWSY/10.png  => JPEG check:  TRUE
Drowsy_datset/train/DROWSY/100.png  => JPEG check:  TRUE
```

We'll have handle BOTH. The following code chunk loads ALL of the images into a drowsy and natural set, by handling png or jpeg (our data cleaning)

We combine the datasets into one train and test dataset.

We look at the initial class imbalance. Notice its not that bad. **Aiden or Jordan, we could use a class imbalance histogram here**

```
    label    n
1  Drowsy 2809
2 Natural 3050

    label   n
1  Drowsy 726
2 Natural 726
```

We shuffle the dataset

Preprocessing

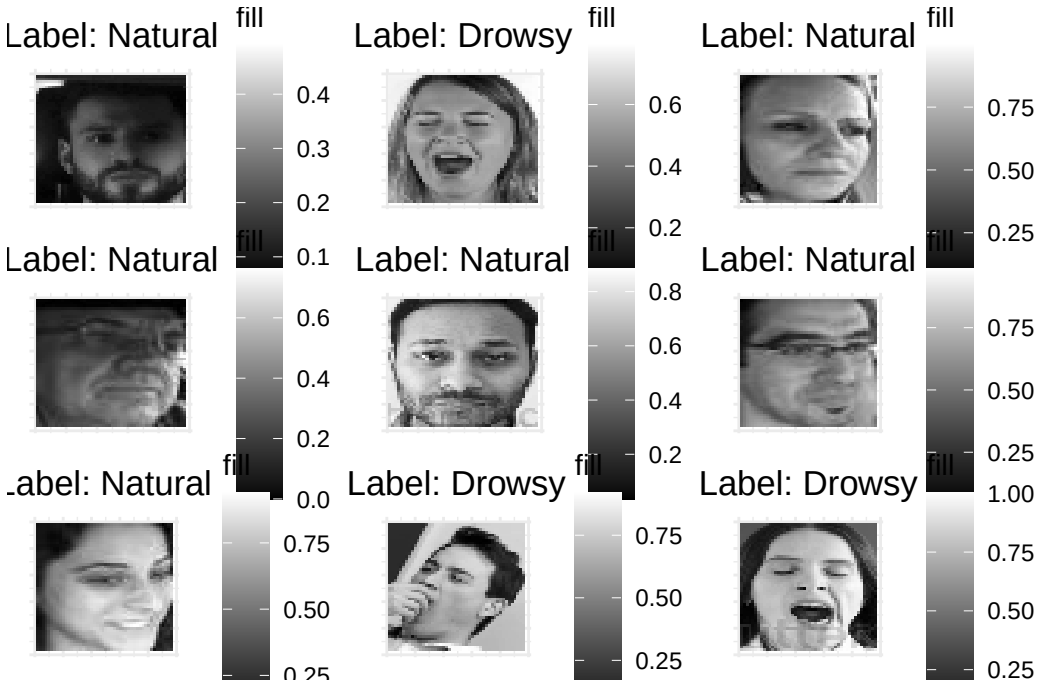
We need to ensure that the datasets are the same going in. This applies CLAHE and HFE to apply pre-processing.

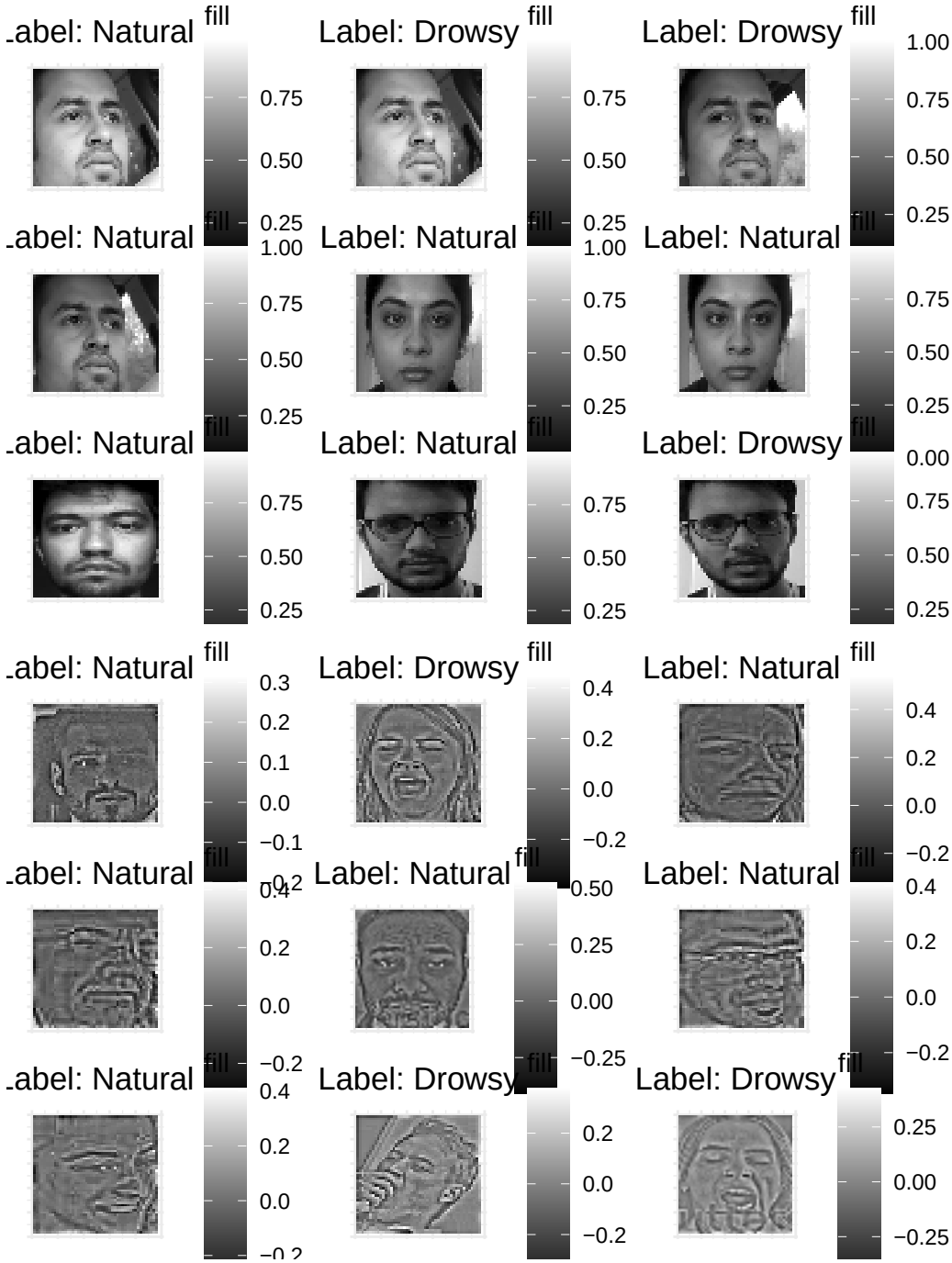
Readying for models

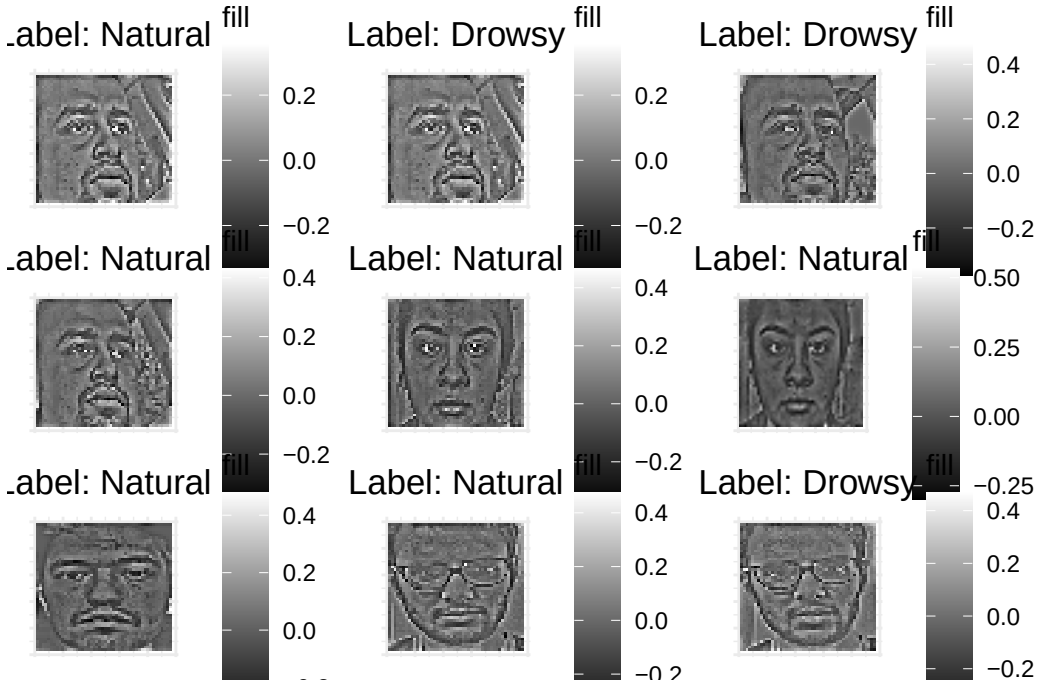
We define our new models as one with feature engineering and one without feature engineering. We stretch and flatten our images into one long string of vectors.

These are plotting functions.

We print the first 9. Notice the differences.




Label: Natural
Label: Drowsy
Label: Drowsy
Label: Natural
Label: Natural
Label: Natural
Label: Natural
Label: Drowsy
Label: Natural
Label: Natural
Label: Natural
Label: Natural
Label: Drowsy
Label: Drowsy



Do PCA

we will do PCA. We will use the PCA done on the train set to transform our test set to then evaluate with our models later.

References

[1] .

[2] Mario Sáenz Espinoza, Ana I Pereira, and José Gonçalves. "Optimization methods for hyper-redundant robots' inverse kinematics in biomedical applications". In: *AIP Conference Proceedings*. Vol. 1479. 2012, p. 818.

[3] Mario Sáenz Espinoza et al. "Inverse Kinematics of a 10 DOF modular hyper-redundant robot resorting to exhaustive and error-optimization methods: A comparative study". In: *Robotics Symposium and Latin American Robotics Symposium (SBR-LARS), 2012 Brazilian*. IEEE. 2012, pp. 125–130.

[4] Ming Tham. *Writing Research Theses or Dissertations*. University of Newcastle Upon Tyne. May 2001. URL: http://lorien.ncl.ac.uk/ming/dept/Tips/writing/thesis/thesis-intro.htm.